Application Architecture & Dependency Record

Fictional training evidence - CR-2026-047

System	Customer Portal
Primary Database	CustomerDB
Architecture Owner	Enterprise Applications Architecture
Last Reviewed	July 2026

Logical Service Path

Customer Portal -> Customer API Service -> CustomerDB

Connected Components

Component	Relationship	Criticality	Notes
Customer Portal	Primary user-facing service	Critical	Requires Customer API and CustomerDB.
Customer API Service	Application integration layer	Critical	Maintains pooled database connections to CustomerDB.
CustomerDB	System of record	Critical	Target of CR-2026-047.
Customer Reporting Service	Downstream read dependency	Medium	Nightly and near-real-time extracts read from CustomerDB.
Notification Service	Indirect dependency	Medium	Receives transaction events generated after successful Customer Portal activity.

Dependency Considerations

- Customer API Service must reconnect successfully after database services restart.
- Customer Reporting Service reads production data from CustomerDB and should be checked after database maintenance.
- Notification Service is not directly changed, but transaction flow should be observed after business validation.

Architecture Constraints

The production Customer API Service uses a persistent connection pool. Database restarts or authentication changes can leave stale connections until the pool is recycled.

Architecture Validation Expectations

- Confirm Customer API Service reconnects to CustomerDB.
- Confirm Customer Portal transaction completes end-to-end.
- Confirm Customer Reporting Service can query CustomerDB.